Victor D. Masters

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Professional Summary

Software Engineer and Systems Architect with 15+ years of equivalent experience delivering high-performance, resilient software solutions across embedded IoT, full-stack platforms, and AI/ML pipelines. Proven expertise in Python, C/C++, and Java, with a strong command of networking fundamentals, TCP/IP protocols, and production debugging in complex environments. Architected scalable SCADA/telemetry systems that quadrupled production output and engineered cloud-native IoT ecosystems using Docker, GitHub Actions, and PostgreSQL. Adept at applying Agile methodologies, CI/CD practices, and Machine Learning techniques to drive operational efficiency, reduce downtime by 75%, and influence technology stack direction.

Professional Experience

Ohio Valley Energy Systems | Austintown, OH Software Engineer & Production Automation Lead | January 2008 – Present * Architected and deployed two full-stack SCADA/telemetry platforms using Python, C, and embedded Linux, quadrupling oil & gas production and generating a $23M revenue increase over two years. * Engineered real-time data pipelines and database architectures (SQL, PostgreSQL) to aggregate field sensor data, enabling predictive analytics that reduced production downtime from 45 days to under 10 days annually. * Implemented robust network protocols and secure communication channels (TLS, mDNS, BLE) across distributed IoT devices, ensuring reliable data transmission and system observability in remote field environments. * Developed automated testing suites and System Acceptance Testing (SAT) protocols, reducing firmware bug rates by 10% and standardizing maintenance workflows across 100+ assets. * Led cross-functional teams using Agile principles to deliver automation hardware and software solutions, conducting code reviews and technical training to elevate team competency. * Designed data-mining visualization tools to identify high-value drilling opportunities, achieving up to 20x production improvements over competitors for four consecutive years.

Independent Software Engineer | Self-Directed Systems Architect & Full-Stack Developer | January 2013 – Present * Designed and built a full-stack IoT aeroponics control platform featuring ESP32 firmware (C/FreeRTOS), cloud microservices (Node.js/Express/PostgreSQL), and React dashboards, utilizing Docker containers and GitHub Actions for CI/CD automation. * Architected a generic-core binary firmware platform for ESP32 devices, enabling a single compiled binary to run across 5+ hardware variants via runtime configuration, eliminating per-variant build matrices and accelerating deployment cycles. * Engineered a 6-stage AI video production pipeline using Python, PyTorch, and LLMs (Qwen3.5-27B) on multi-GPU infrastructure (3x RTX 5090), implementing lazy-loading model managers that reduced inference load times by 90%. * Developed reusable ESP-IDF component libraries for networking, telemetry, and control, shared across multiple projects to reduce code duplication and improve system reliability and maintainability. * Implemented end-to-end security protocols including SRP6a secure pairing, HTTPS/TLS command channels, and mDNS zero-configuration discovery, ensuring secure device onboarding and network resilience. * Built autonomous dual-hotend 3D printer controller with CAN-bus communication and independent PID temperature control, demonstrating expertise in real-time systems and motion control algorithms.

Technical Skills

SCADA Systems, Telemetry, Real-Time Data Pipelines, Database Architecture, System Integration, API Development, Data Visualization, Automation Controls, Project Management, CI/CD Pipelines, GitHub Actions, Docker, Git, Agile, Scrum, Code Reviews, Automated Testing, System Acceptance Testing (SAT), Benchmark Testing, System Architecture, Scalability Design, Production Debugging, TCP/IP Stack, Network Protocols, Routing, mDNS, BLE 5.0, TLS/HTTPS, CAN-bus, Ethernet, Embedded Linux, ESP-IDF, FreeRTOS, Machine Learning, LLMs, Diffusion Models, Multi-GPU Inference, Data Modeling, Configuration Management, Software Documentation, Technical Troubleshooting, Cross-functional Collaboration

Technologies

Python, C, C++, Java, Kotlin, SQL, TypeScript, Bash, Ruby, Perl, PyTorch, Node.js, Express, React, Jetpack Compose, Diffusers, vLLM, PostgreSQL, MySQL, SQLite, MS SQL Server, Linux, Windows, UNIX, Android, ESP32, PIC, PLC, ARM, EFR32

Key Achievements

Quadrupled oil & gas production and generated $23M in revenue increase by architecting custom SCADA/telemetry platforms and data mining tools.

Reduced production downtime by 75% (from 45 days to <10 days annually) through implementation of automated monitoring, predictive analytics, and standardized maintenance protocols.

Engineered a generic-core firmware architecture deployed across 5+ IoT device variants, eliminating build matrix complexity and accelerating time-to-market.

Built a high-performance AI inference pipeline on 96GB local GPU VRAM with zero cloud dependencies, reducing model load times by 90% via lazy-loading optimization.

Achieved 20x production improvements over competitors for 4 years through process analysis and engineering innovations.

Education

Coursework in Civil Engineering, Construction Management, HVAC/R, Industrial Electronics | Columbus State Community College | 1997 – 2002